

Bridging the Inference Gap: Consistent Training for Low-Resource Cross-Lingual Intervention

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Introduction

Language Models perform well overall but remain heavily English-centric

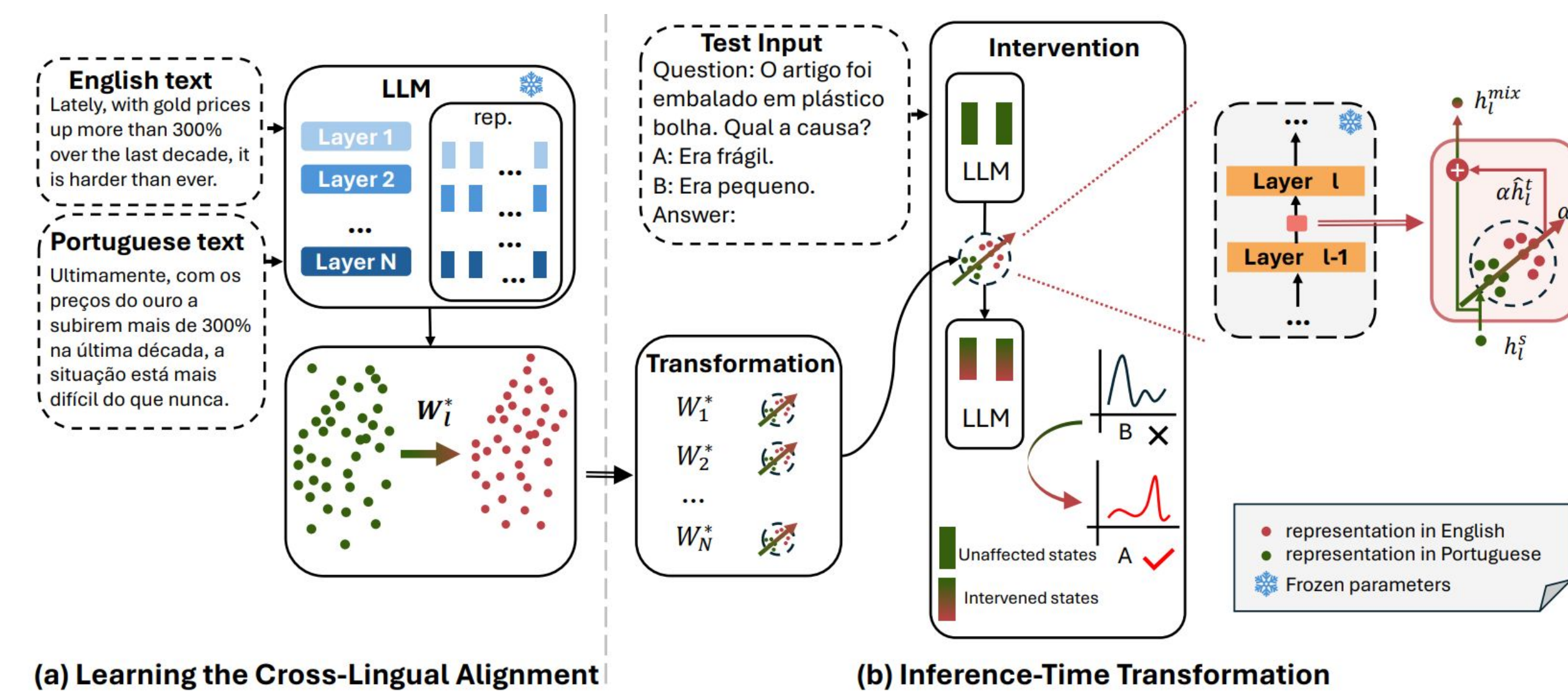
- Large performance gaps across low-resource languages
- Fine-tuning and retraining are costly

Question: *Can we reduce cross-lingual gaps without retraining?*

We propose INCLINE : Inference-Time Cross-Lingual Intervention

- Aligns internal representations across languages
- Transfers knowledge from high- to low-performing languages
- Applied only at inference time with no parameter updates

Methodology



Training: Learning Alignment

We learn a matrix W_l^* to map source language representations (h^s) to the target language space (h^t) by minimizing the distance between them

$$W_l^* = \underset{W_l}{\operatorname{argmin}} \sum_{i=1}^N ||W_l h_{i,l}^s - h_{i,l}^t||^2$$

Inference: Transformation & Intervention

During inference, we project the test input ($h_{q,l}^s$) into the target space and blend it with the original representation using α , a strength hyperparameter

$$\begin{aligned} \text{Projection: } \hat{h}_{q,l}^t &= W_l^* h_{q,l}^s \\ \text{Intervention: } h_{q,l}^{mix} &= \hat{h}_{q,l}^t + \alpha \hat{h}_{q,l}^s \end{aligned}$$

Where,
 l is Layer index
 h^s/h^t is Hidden states of source/target

Limitations

- Over-reliance on XStoryCloze:** Most experiments focus on XStoryCloze (a discriminative task), while limited discussion is provided on generative tasks like Translation.
- Focus on High-Resource (X) to English:** While the method has potential for low-resource languages, only high-resource (X) to (English) translation was considered.
- Reliance on BLEU:** Evaluation is limited to BLEU scores even though LLMs tend to paraphrase significantly.

Improved methodology

Proposed Improvement: Consistency

INCLINE trains W for direct projection ($W h^s \approx h^t$) but uses it for mixing ($h^s + \alpha W h^s$) during inference, creating an objective mismatch.

We modify the training objective to directly optimize the post-intervention vector, ensuring W learns the specific shift needed given the mixing factor α .

New Training Objective:

$$W_l^* = \underset{W_l}{\operatorname{argmin}} \sum_{i=1}^N ||(h_{i,l}^s + \alpha W_l h_{i,l}^s) - h_{i,l}^t||^2$$

Implementation details

- We start with Llama-3-8B-Instruct as our base LLM.
- We focus exclusively on the **low-resource translation** task.
- All experiments use **quantized** model for viable compute.
- Sentence representations are extracted from the **last token** of both source and target sentences.
- We use L2 regularization since the system is highly over parameterized.
- Translation quality is evaluated using **BLEU** and **ChrF++**. For African languages, we additionally report **AfriCOMET**.

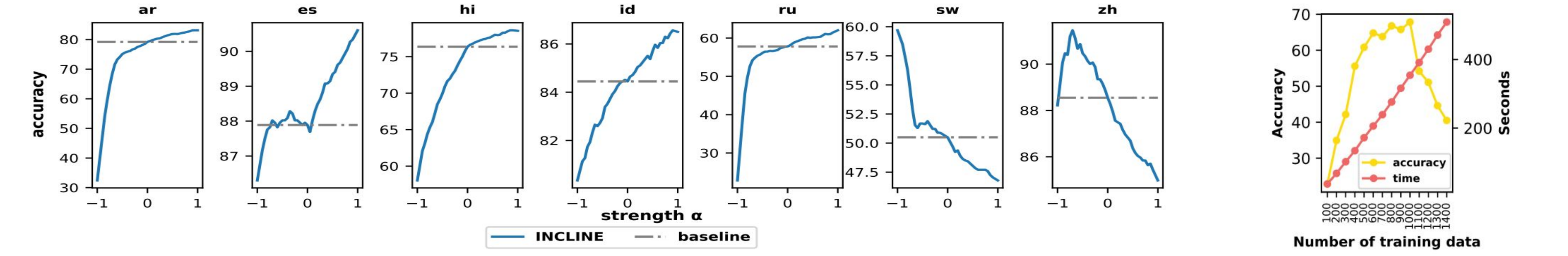
Evaluated Approaches

- No intervention** uses the quantized base model for translation, with no alignment or model modification.
- Intervention** applies the learned alignment matrices at inference time, modifying hidden states using a fixed intervention strength (referred to as INCLINE).
- Consistent intervention** applies the proposed improvement in training the alignment matrices and applies these alignment matrices in the same way during inference (referred to as Improved INCLINE).
- LoRA** trains low-rank adapters on the same FLORES translation data. Since both LoRA fine-tuning and INCLINE adds additional intermediate weights, by comparing both, we gauge into the effectiveness of the indirect INCLINE objective against the more apparent token prediction loss.

Takeaway

- INCLINE performs well even in severe low-resource settings.
- LoRA significantly outperforms INCLINE, but INCLINE offers a much quicker and more efficient way to improve base model.
- Improvement is consistent across scripts, though better for Latin scripts (e.g., Serbian Latin vs. Cyrillic).
- INCLINE shows improvements even for languages with very sparse tokenization (Lao & Burmese).
- Our proposed INCLINE improvement achieves better performance for African Languages while matching performance for others.
- The regularized objective is stable, unlike the non-regularized approach which failed on Sanskrit.

Relevant existing experiments



WMT23	en	de	ja	ru	uk	zh
BASELINE	-	18.26	10.17	14.73	11.36	14.39
INCLINE	-	18.85 [†]	10.30 [†]	15.24 [†]	11.71 [†]	15.05 [†]

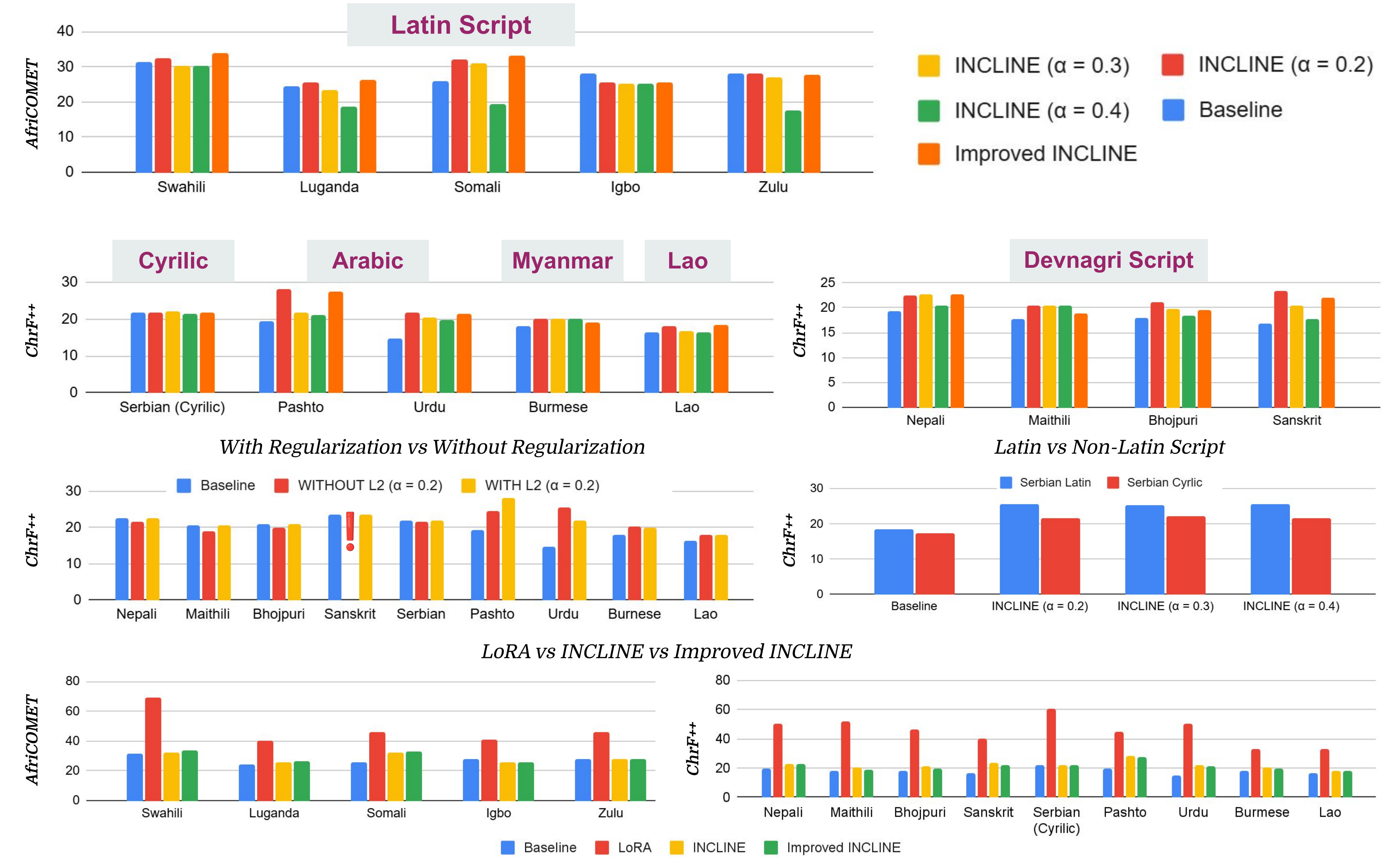


Notice that **NONE** of the languages considered for translation are low-resource

Our experiments

Questions we seek to answer:

- Whether INCLINE works well for low resource languages
- Whether performance of INCLINE is dependent on the script of the high resource language
- Whether tokenization density affects the performance.



References

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